

Elementary Problems

Operation, Game and Invariance

Version 1.0

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Basic Work

1. There is a row of n light bulbs labelled from 1 to n . Each light bulb can be either 'on' or 'off'. Each time one may change the states (from 'on' to 'off' or from 'off' to 'on') of any two consecutive light bulbs simultaneously, or change the state of the light bulb with label 1. Prove that for any initial states of the light bulbs, it is always possible to make all light bulbs 'off' through some steps.
2. There are $n \geq 2$ cards lying in a row, each having a white side and a black side. In each step, we randomly choose at least 2 consecutive cards where the 2 cards at the two ends have their white sides facing upwards and all cards in between have their black sides facing upwards, and we flip each of these cards so that the colour of each card changes. Is it always true that the process must come to an end after finitely many steps (meaning that we can no longer choose such a group of cards) regardless of the initial configuration and the choice we make in each step?
3. (IMO shortlist 1977) There are 2^n words of length n over the alphabet $\{0, 1\}$. Prove that the following algorithm generates the sequence $w_0, w_1, \dots, w_{2^n-1}$ of all these words such that any two consecutive words differ in exactly one digit.
 - (1) $w_0 = 00 \cdots 0$ (n zeros).
 - (2) Suppose $w_{m-1} = a_1 a_2 \cdots a_n$, $a_i \in \{0, 1\}$. Let $e(m)$ be the exponent of 2 in the representation of m as a product of primes, and let $j = 1 + e(m)$. Replace the digit a_j in the word w_{m-1} by $1 - a_j$. The obtained word is w_m .
4. (Russian MO 2010 Grade 11 Problem 2) On an $n \times n$ chart, where $n \geq 4$, stand '+' signs in the cells of the main diagonal and '-' signs in all the other cells. You can change all the signs in one row or in one column, from - to + or from + to -. Prove that you will always have n or more + signs after finitely many operations.
5. X and Y play the following game. Firstly, X chooses a number n from the first 100 integers secretly. Y needs to ask questions to deduce the value of n . Each time Y can choose any positive integers a and b , and ask, 'is $n + a$ a multiple of b ?' Then X will give a correct response, either 'yes' or 'no'. Afterwards, Y can ask the next question and so on.

What is the minimum number of questions that Y needs to ask so that Y must be able to deduce the value of n ?
6. (IMO shortlist 1994 C1) Two players play alternately on a 5×5 board. The first player always enters a 1 into an empty square and the second player always enters a 0 into an empty square. When the board is full, the sum of the numbers in each of the nine 3×3 squares is calculated and the first player's score is the largest such sum. What is the largest score the first player can make, regardless of the responses of the second player?

7. (Russian MO 2011 Grade 11 Problem 6) There are more than n^2 stones on the table. Peter and Vasya play a game, starting from Peter. Each turn a player can take any prime number less than n stones, or any multiple of n stones, or 1 stone. Prove that Peter can always take the last stone (regardless of Vasya's strategy).

Algorithm

8. (IMO shortlist 1966) Consider n students with numbers $1, 2, \dots, n$ standing in the order $1, 2, \dots, n$. Upon a command, any of the students either remains on his place or switches his place with another student. (Actually, if student A switches his place with student B , then B cannot switch his place with any other student C any more until the next command comes.)

Is it possible to arrange the students in the order $n, 1, 2, \dots, n-1$ after two commands?

9. Starting from X , players X and Y alternately choose a number from the first 99 positive integers which has not been chosen by anyone. The rule is that the number chosen by any player and the number just chosen by the opponent has to be relatively prime (where there is no restriction on the first number chosen by X). The one who cannot do so will lose. Who has a winning strategy?
10. Starting from X , players X and Y alternately choose a number from the first 30 positive integers which has not been chosen by anyone. The rule is that the sum of the number chosen by any player and the number just chosen by the opponent has to be a prime number (where there is no restriction on the first number chosen by X). The one who cannot do so will lose. Who has a winning strategy?
11. Two players X and Y play a game as follows. During the game, player X always uses red colour while player Y always uses blue colour. In round 1, starting from player X , they take turns marking a point on a given circle which has not been marked. Round 1 ends when each of them has marked 2017 points on the circle. In round 2, starting from player X , they take turns joining a chord using two of the marked points such that the endpoints of each chord have the same colour as that chord, and any two chords which intersect in the interior of the circle must have the same colour. Each chord can only be joined once. The game ends when one of the players cannot perform the operation, and the other player will win the game. Who has a winning strategy?
12. There are 1234 cards labelled from 1 to 1234. Starting from X , players X and Y alternately choose one card from the cards which have not been chosen. When all the cards have been chosen, the unique player whose cards have labels with an even sum wins. Who has a winning strategy?

13. (IMO shortlist 1985) Let m boxes be given, with some balls in each box. Let $n < m$ be a given integer. The following operation is performed: choose n of the boxes and put 1 ball in each of them. Prove:
 - (a) If m and n are relatively prime, then it is possible, by performing the operation a finite number of times, to arrive at the situation that all the boxes contain an equal number of balls.
14. (Tournament of Towns 2001) Two persons play a game on a board divided into 3×100 squares. They move in turn: the first places tiles of size 1×2 lengthwise (along the long axis of the board), the second, in the perpendicular direction. The loser is the one who cannot make a move. Which of the players can always win (no matter how his opponent plays), and what is the winning strategy?
15. (Japan MO Finals 2011 Problem 3) Person A writes down nonnegative integers in each N grid running in a line horizontally. When A says one nonnegative integer, person B replaces some number in N grid by the number that A said. Repeat this procedure. When these numbers are arranged in the order of monotone increasing in the wider sense, the procedure is over. Is it possible that B can finish in regardless of A ?
16. (Russian MO 2004 Grade 9 Problem 3) There are 2004 boxes on a table, each of them containing a ball. We know that some of the balls are white and that their number is even. We are allowed to choose any two boxes and ask whether at least one of them contains a white ball. What is the minimum number of questions that guarantees us to be able to specify some two boxes containing white balls?
17. (Bosnia and Herzegovina TST 2016 Problem 2) Let n be a positive integer and let t be an integer. n distinct integers are written on a table. Bob, sitting in a room nearby, wants to know whether there exist some of these numbers such that their sum is equal to t . Alice is standing in front of the table and she wants to help him. At the beginning, she tells him only the initial sum of all numbers on the table. After that, in every move he says one of the 4 sentences:
 - (i) Is there a number on the table equal to k ?
 - (ii) If a number k exists on the table, erase him.
 - (iii) If a number k does not exist on the table, add him.
 - (iv) Do the numbers written on the table can be arranged in two sets with equal sum of elements?

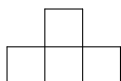
On these questions Alice answers yes or no, and the operations he says to her she does (if it is possible) and does not tell him did she do it. Prove that in less than $3n$ moves, Bob can find out whether there exist numbers initially written on the board such that their sum is equal to t .

Tiling and Colouring

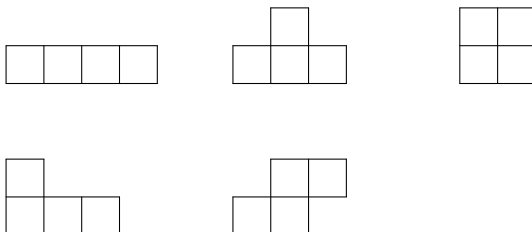
18. Consider an 8×8 table with two opposite corners removed. Prove that it is not possible to tile the remaining 62 cells with 31 dominoes (rotation is allowed).



19. Find all $n \times n$ tables which can be tiled completely with T -tetrominoes (rotation is allowed).



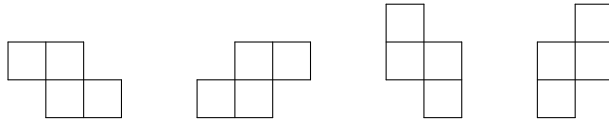
20. Prove that it is not possible to use the following tetrominoes to tile a rectangle so that each tetromino is used exactly once.



21. (Mexico MO 2007 Problem 5) In each square of a 6×6 grid there is a lightning bug on or off. One move is to choose three consecutive squares, either horizontal or vertical, and change the lightning bugs in those 3 squares from off to on or from on to off. Show if at the beginning there is one lightning bug on and the rest of them off, it is not possible to make some moves so that at the end they are all turned off.

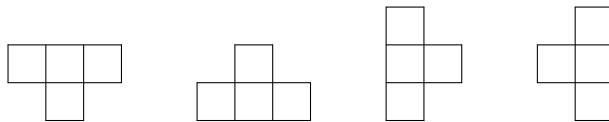
22. (IMO shortlist 1984)

- (a) Decide whether the fields of the 8×8 chessboard can be numbered by the numbers $1, 2, \dots, 64$ in such a way that the sum of the four numbers in each of its parts of one of the forms

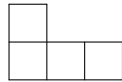


is divisible by four.

- (b) Solve the analogous problem for



23. Find all $n \times n$ tables (where $n \geq 3$) with the 4 corner cells removed which can be tiled completely with L -tetrominoes (rotation and reflection are allowed).



Invariance

24. Let n and k be positive integers. There are n coins in a pile. Players X and Y alternately take away some coins from the pile. Each time one can take away at least 1 and at most k coins. The one who takes away all remaining coins will win. Who has a winning strategy?
25. Let n and k be positive integers. There are n coins in a pile. Players X and Y alternately take away some coins from the pile. Each time one can take away 2^t coins for some $t = 0, 1, 2, \dots$ (the choice of t may vary in each step). The one who takes away all remaining coins will win. Who has a winning strategy?

26. Let n and k be positive integers. There are n coins in a pile. Players X and Y alternately take away some coins from the pile. Each time one can take away 1, 3 or 4 coins. (If there are exactly 2 coins left, one cannot take away both coins.) The one who takes away all remaining coins will win. Who has a winning strategy?
27. (Costa Rica Final Round 2003 Problem 1) Two players A and B participate in the following game. Initially we have a pile of 2003 stones. A plays first, and he picks a divisor of 2003 and removes that number of stones from the pile. Then B picks a divisor of the number of remaining stones, and removes that number of stones from the pile, and so forth. The player who removes the last stone loses. Prove that one of the players has a winning strategy and describe it.
28. Let n be a positive integer. The integers $1, 2, \dots, 4n + 2$ are written on the blackboard. In each step, we erase two integers a and b on the blackboard and replace them by the integer $|a - b|$. We repeat the steps until exactly one integer is left. Prove that this integer cannot be 0.
29. Let n be a positive integer. The expression $+1 + 2 + \dots + n$ is written on the blackboard. We can replace some (possibly all or none) positive signs on the blackboard by negative signs. Then we compute the answer obtained. How many different answers can we obtain?
30. Let n be a positive integer. There are $2n$ piles of coins lying in a circle. We order the piles in clockwise direction. Initially, each of the first and the third pile has 2 coins, while all the other piles have 1 coin each. In each step, we can add 1 coin to each of the two neighbouring piles. Prove that it is not possible to make all piles have the same number of coins after finitely many steps.
31. Let $n \geq 2$ be a positive integer. Consider an $n \times n$ chessboard with 3 black corner cells and 1 white corner cell. We colour each of the remaining cells black or white arbitrarily. Prove that at the end we can always find a 2×2 square of the chessboard which consists of an odd number of black cells.
32. Let n and k be positive integers with $k \leq n$. There are n cards each having a white side and a black side. Initially, all cards have their white sides facing upwards. In each step, we can flip any k cards so that the colour of each card changes. Find all n and k such that it is possible to make all cards have their black sides facing upwards after finitely many steps.
33. (Iran MO 3rd Round 2017) There are 100 points on the circumference of a circle, arbitrarily labelled by $1, 2, \dots, 100$. For each three points, call their triangle clockwise if the increasing order of them is in clockwise order. Prove that it is impossible to have exactly 2017 clockwise triangles.

34. (CWMO 2008 Problem 5) Four frogs are positioned at four points on a straight line such that the distance between any two neighbouring points is 1 unit length. Suppose every frog can jump to its corresponding point of reflection, by taking any one of the other 3 frogs as the reference point. Prove that, there is no such case that the distance between any two neighbouring points, where the frogs stay, are all equal to 2008 unit length.
35. (IMO shortlist 1989) A natural number is written in each square of an $m \times n$ chessboard. The allowed move is to add an integer k to each of two adjacent numbers in such a way that nonnegative numbers are obtained. (Two squares are adjacent if they have a common side.) Find a necessary and sufficient condition for it to be possible for all the numbers to be zero after finitely many operations.
36. (Macedonia MO 2010 Problem 3) A total of 2010 coins are distributed in 5 boxes. At the beginning the quantities of coins in the boxes are consecutive natural numbers. Martha should choose and take one of the boxes, but before that she can do the following transformation finitely many times: from a box with at least 4 coins she can transfer one coin to each of the other boxes.

What is the maximum number of coins that Martha can take away?

(Remark: This means the coins in the 5 boxes are 400, 401, 402, 403, 404 respectively at the beginning.)

37. (IMO shortlist 1985) Let m boxes be given, with some balls in each box. Let $n < m$ be a given integer. The following operation is performed: choose n of the boxes and put 1 ball in each of them. Prove:
- (b) If m and n are not relatively prime, there exist initial distributions of balls in the boxes such that an equal distribution is not possible to achieve.

(Remark: as indicated in (a), this means the situation that all the boxes contain an equal number of balls.)

38. Let n be a positive integer. Prove that no matter how we partition the set $\{1, 2, \dots, 2n\}$ into two sets $A = \{a_1, a_2, \dots, a_n\}$ and $B = \{b_1, b_2, \dots, b_n\}$ such that $a_1 < a_2 < \dots < a_n$ and $b_1 > b_2 > \dots > b_n$, we have

$$|a_1 - b_1| + |a_2 - b_2| + \dots + |a_n - b_n| = n^2.$$

39. (European Mathematical Cup 2016 Senior Category Problem 2) For two positive integers a and b , Ivica and Marica play the following game: Given two piles of a and b cookies, on each turn a player takes $2n$ cookies from one of the piles, of which he eats n and puts n of them on the other pile. Number n is arbitrary in every move. Players take turns alternatively, with Ivica going first. The player who cannot make a move, loses. Assuming both players play perfectly, determine all pairs of numbers (a, b) for which Marica has a winning strategy.

40. (Balkan MO 2018 Problem 3) Alice and Bob play the following game: They start with nonempty piles of coins. Taking turns, with Alice playing first, each player chooses a pile with an even number of coins and moves half of the coins of this pile to the other pile. The game ends if a player cannot move, in which case the other player wins. Determine all pairs (a, b) of positive integers such that if initially the two piles have a and b coins respectively, then Bob has a winning strategy.
41. There are n copies of 1's written on the blackboard. Each time we can erase two numbers a and b on the blackboard and write down the number $\frac{a+b}{4}$. After $n-1$ steps, exactly one number remains. Prove that the final number is at least $\frac{1}{n}$.
42. (ELMO shortlist 2010 C4) The numbers $1, 2, \dots, n$ are written on a blackboard. Each minute, a student goes up to the board, chooses two numbers x and y , erases them, and writes the number $2x+2y$ on the board. This continues until only one number remains. Prove that this number is at least $\frac{4}{9}n^3$.

Record

v1.0: problems 1–42